Lesson 21: Survey Research: Constructing Surveys

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 7.1: Prelude to Survey Research; 7.2: Overview of Survey Research; 7.3: Constructing Surveys.

In this lesson

Survey research measures thoughts, feelings and behaviors through self-reports from carefully selected samples. You will learn what defines a survey, why answering a question is a psychological process open to context effects, and how to write good items with the BRUSO model. In English, you will practice participle clauses that interpret results and the precise question forms that good items need.

Learning goals

After this lesson you should be able to:

- Define survey research by its two characteristics and explain how a survey can also be an experiment.
- Describe the five steps of survey responding and two kinds of context effect.
- Choose between open-ended and closed-ended items and evaluate items with BRUSO.
- Build balanced rating scales and mutually exclusive, exhaustive categories, and order a questionnaire.
- Interpret results with participle clauses (*suggesting that*, *showing how*).

Key language

Term or phrase	Meaning	Example
survey research	measuring variables with self-reports, with close attention to sampling	the National Comorbidity Survey
respondents	the usual term for participants in survey research	nearly 2,000 respondents aged 13 to 88
context effect	an unintended effect of surrounding items or the response scale on responses to an item	middle options read as "normal"
item-order effect	the effect of responding to one item on responses to a later item	dating frequency asked before life satisfaction
open-ended item	asks a question and lets respondents answer however they want	"Is there anything else you would like to tell us about?"
closed-ended item	asks a question and provides response options to choose from	"How old are you?" with age ranges

Term or phrase	Meaning	Example
rating scale	an ordered set of response options to a closed-ended item	Never, Rarely, Sometimes, Often, Always
Likert scale	agreement ratings (5 points) on several favorable and unfavorable statements, summed	an attitude toward a group
BRUSO	a model for good items: brief, relevant, unambiguous, specific, objective	"Have you ever owned a gun?"
double-barrelled item	asks about two separate issues but allows only one response	"...feeling anxious and depressed"
mutually exclusive / exhaustive	categories that do not overlap / that cover all possible responses	*Protestant* and *Catholic* (exclusive, not exhaustive)
suggesting that	introduces an interpretation of a result	"...only –.12, suggesting that the two variables are only weakly related"
At best ... At worst ...	the mildest and the most serious outcome	"At best, these influences add noise to the data. At worst, they result in systematic biases"

Rules and explanations

The ideas

What survey research is. Two characteristics: (1) variables are measured with *self-reports* (questionnaires or interviews); (2) sampling gets close attention, with a strong preference for large random samples, which give the most accurate estimates of what is true in the population. Beyond these, "almost anything goes" in length, mode and topic. Most surveys are non-experimental: they describe single variables or relationships (e.g., income and health). But Lerner et al. (2003) surveyed nearly 2,000 Americans aged 13 to 88 online after the September 2001 attacks and found that those primed to feel anger perceived less risk than those primed to feel fear. Self-reports and a large national sample make this survey research; manipulating emotion makes it an experiment.

History and uses. Surveys grew from social surveys of poverty, government surveys, market research and election polling. In 1936, *Literary Digest* sent ballots to millions and predicted a landslide for Landon; Gallup, with much smaller scientific samples, predicted Roosevelt's victory, and was right. In psychology, surveys are tied to the study of attitudes, stereotypes and prejudice, and they estimate the lifetime prevalence of disorders (the National Comorbidity Survey interviewed nearly 10,000 adults at home in 2002 and 2003).

Answering is a cognitive process (Sudman, Bradburn & Schwarz, 1996). Respondents (1) **interpret** the question, (2) **retrieve** information from memory, (3) form a tentative **judgment**, (4) **format** it into a response option and (5) **edit** the response if necessary. The item "How many alcoholic drinks do you consume in a typical day?" (from "a lot more than average" to "a lot fewer than

average") causes problems at every step: Does beer count? Is a "typical day" a weekday? What is "average"? A heavy drinker may edit the answer to avoid looking bad.

Context effects come from an item's context, not its content:

- *Item order*: when life satisfaction was rated first, its correlation with dating frequency was −.12; when dating frequency came first, it was +.66, because dating was then more accessible in memory (Strack et al., 1988).
- *Response options*: a scale from "less than once a year" to "more than once a month" makes people think of major irritations and report few; people also treat the middle option as normal, reporting more television when it is 4 hours than when it is 2.
- *Remedy*: when there is no natural order, rotate, counterbalance or randomize items and options.

Open-ended or closed-ended?

	Open-ended	Closed-ended
Use when	answers are unknown, influence must be avoided, or the question is still vague	likely answers are known; the variable is well defined
Researcher	easy to write; hard to analyze	harder to write; easy to analyze
Respondent	slower; often skipped	quick and easy

Rating scales have 3 to 11 options, most often 5 or 7: 5 for *unipolar* scales (one construct, e.g., frequency: Never ... Always), 7 for *bipolar* scales (Like very much ... Dislike very much), where *branching* ("Do you generally like or dislike ice cream?" first) improves reliability and validity. Show verbal labels; convert them to numbers in the analysis. A *Likert scale*, strictly, sums agreement ratings on several favorable and unfavorable statements; a single 0-to-10 item is just a rating scale.

BRUSO (Peterson, 2000):

Criterion	Poor item	Effective item
Brief	"Are you now or have you ever been the possessor of a firearm?"	"Have you ever owned a gun?"
Relevant	"What is your sexual orientation?"	Leave it out unless clearly relevant.
Unambiguous	"Are you a gun person?"	"Do you currently own a gun?"
Specific	"How much have you read about the new gun control measure and sales tax?"	"How much have you read about the new sales tax?"
Objective	"How much do you support the new gun control measure?"	"What is your view of the new gun control measure?"

A *pilot test* shows how people interpret an item: ask a few people to explain how they understood it.

Response options. Categories should be *mutually exclusive* (*Christian* and *Catholic* overlap) and *exhaustive* (add *Other*, with space to write in). Balance the extremes around a neutral midpoint: *Extremely Unlikely* | *Somewhat Unlikely* | *As Likely as Not* | *Somewhat Likely* | *Extremely Likely*. A

middle option is optional: leaving it out encourages people to think more deeply; including it lets them choose "neither."

Formatting. An *introduction* encourages participation (purpose, sponsor, incentives) and establishes informed consent (topics, time, the right to withdraw, confidentiality). Then come clear *instructions*, the *most important items* (while respondents are most interested and least tired), items grouped by topic or type, *demographic items last*, and *thanks*.

Language focus

1. Interpreting a result with a participle clause (comma + *-ing* clause):

- "When the life satisfaction item came first, the correlation between the two was only –.12, *suggesting that* the two variables are only weakly related."
- "...the participants who were primed to feel anger perceived less risk than the participants who had been primed to feel fear—*showing how* risk perceptions are strongly tied to specific emotions."

The verb sets the strength of the claim: *suggesting* (tentative) or *showing*, *demonstrating* (strong).

2. Plain, precise questions. BRUSO is partly a language rule:

- Prefer a verb to a heavy noun phrase: "be the possessor of a firearm" becomes "own a gun."
- Use question word order (auxiliary + subject + verb): "*Do you* currently own a gun?"
- One question, one issue: *and* in an item may signal a double-barrelled item.
- Neutral wording: "What is your view of ...?", not "How much do you support ...?"

> **Note for Arabic speakers** - Arabic questions can keep statement word order, but English needs an auxiliary before the subject: "How many hours *do you* sleep?", not "How many hours you sleep?" - For experience up to now, Arabic uses the past tense, while English normally uses the present perfect, as in the book's item "*Have you ever* in your adult life *been* depressed for a period of 2 weeks or more?"

Practice

1. Reading questions

- 1.1 What are the two characteristics of survey research?
- 1.2 Why is the study by Lerner et al. (2003) both survey research and an experiment?
- 1.3 In the National Comorbidity Survey table, which disorder has the highest total lifetime prevalence? For which disorders is prevalence higher in males?
- 1.4 Why did the correlation between dating frequency and life satisfaction change from –.12 to +.66?
- 1.5 Give two disadvantages of open-ended items.
- 1.6 What are the two functions of a survey introduction, and why are demographic items often last?

2. Applying the section (Sections 7.1-7.3 have no exercises; the chapter's exercises are in Lesson 22.)

- 2.1 Rewrite the alcohol item to solve the problems of interpretation and response options that the section names.
- 2.2 Which BRUSO criterion does "Please rate the extent to which you have been feeling anxious and depressed" break? Fix it.
- 2.3 A religion item offers *Christian*, *Catholic*, *Protestant*. What is wrong? Fix it.
- 2.4 Balance this scale: *Unlikely* | *Somewhat Likely* | *Likely* | *Very Likely* | *Extremely Likely*.

3. Language. Add an interpretation to each result with a comma and an *-ing* clause. Would *showing* be better than *suggesting*?

- (a) When the dating frequency item came first, the correlation between the two was +.66. (Interpretation: those who date more have a strong tendency to be more satisfied with their lives.)
- (b) People reported watching more television when the middle option was 4 hours than when it was 2 hours. (Interpretation: they treat the middle option as typical.)

Speak with your tutor

- Choose a survey item you have answered recently. Walk the tutor through the five steps of responding to it, and ask how the item could be improved.
- Pilot test: give the tutor three items from your writing task and ask it to explain how it interprets each. Revise any word it reads differently.
- In PSYC 110 Lecture 20 (The Good Life: Happiness), Professor Bloom describes phone interviews in which people rated their lives as happier on sunny days, an effect that disappeared when they were first asked "How's the weather?" Explain this as a context effect and compare it with Strack et al. (1988).
- In PSYC 110 Lecture 9 (Love), Professor Salovey reports that students in surveys say looks are unimportant, yet after blind dates, looks best predicted who met again. Which step of survey responding might explain this gap?

Writing task

Research project: survey items, first draft (250-350 words). Draft a short questionnaire for your study of sleep and memory in university students; you will pilot and finish it in Lesson 22.

1. An introduction (60-90 words): purpose, sponsor (e.g., your university), time needed, the right to withdraw, confidentiality.
2. Instructions, with an example for any unusual scale.
3. Eight items, the most important first: at least four closed-ended items, including a 5-point frequency scale (e.g., how often students feel sleepy in morning classes) and a balanced scale; one categorical item; one open-ended item; demographic items at the end.
4. A thank-you.
5. One sentence on how you will avoid an item-order effect between sleep items and memory items.

Model item: "On a typical weeknight (Sunday to Thursday), about how many hours do you sleep? ____ hours." If you work on your own topic, write items for its key variables.

Self-editing checklist

- Is every item brief, relevant, unambiguous, specific and objective?
- Does any *and* make an item double-barrelled?
- Are time frames and key words defined?
- Are categories mutually exclusive and exhaustive?
- Are rating scales balanced, with verbal labels?
- Does the introduction cover informed consent?
- Does every question use correct question word order?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- The xkcd comic "Question" printed in Section 7.3 (a closed-ended item whose respondent invents his own answer): https://imgs.xkcd.com/comics/question.png

Answer key

1. Reading questions

- 1.1 Variables are measured with self-reports (questionnaires or interviews), and close attention is paid to sampling, with a strong preference for large random samples (7.2, first paragraph).
- 1.2 It used self-report measures and a large national sample (survey research) and manipulated an independent variable, anger vs. fear, to assess its effect on risk judgments (experiment).
- 1.3 Major depressive disorder (16.9%). Higher in males: alcohol abuse (19.6% vs. 7.5%) and drug abuse (11.6% vs. 4.8%).
- 1.4 Reporting dating frequency first made that information more accessible in memory, so respondents based their life satisfaction rating on it: an item-order effect.
- 1.5 Any two: they take more time and effort; they are harder to analyze (answers must be transcribed and coded); respondents skip them more often.
- 1.6 To encourage participation and to establish informed consent. Demographic items are least interesting but easy to answer when respondents are tired or bored.

2. Applying the section

- 2.1 (example answer) "On a typical weekday (Monday to Friday), how many alcoholic drinks (beer, wine or spirits) do you have? ____ drinks," plus the same item for a typical weekend day. Defined terms remove the ambiguity, and a number replaces the vague "average" options. Other good versions are possible.
- 2.2 *Specific*: it is double-barrelled. Write one item about feeling anxious and one about feeling depressed.
- 2.3 *Christian* and *Catholic* overlap (not mutually exclusive), and the list is not exhaustive. Use non-overlapping categories (*Protestant*, *Catholic*, *Jewish*, *Hindu*, *Buddhist* ...) plus *Other (please specify):* _____.
- 2.4 *Extremely Unlikely* | *Somewhat Unlikely* | *As Likely as Not* | *Somewhat Likely* | *Extremely Likely* (the book's version).

3. Language

- (a) "..., suggesting that those who date more have a strong tendency to be more satisfied with their lives" (the book's sentence). (b) "..., suggesting that they treat the middle option as typical."
- *Suggesting* is better: each result fits the interpretation but does not prove it. In (a), the +.66 was itself produced partly by item order, so a strong claim would be risky.

My notes